

Confidence intervals, t-score

Background You are given the dataset from the lesson

Task 1 Calculate the mean and the standard error of the dataset

Task 2 Determine which statistic to use for inference

Task 3 Find the appropriate statistic, taking into consideration the degrees of freedom (if applicable) for 99% confidence

Task 4 Find the 99% confidence interval

Dataset

\$ 78,000
\$ 90,000
\$ 75,000
\$ 117,000
\$ 105,000
\$ 96,000
\$ 89,500
\$ 102,300
\$ 80,000

Mean	\$ 92,533
Standard Error	\$ 4,644

Standard deviation	\$ 13,932
Square root of N	3

Statistic	t-statistic
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99% Confidence level

α	0.01
$\alpha/2$	0.005
$(1-\alpha/2)$	0.995

99% CI	3.355
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Note: A **confidence interval** is an interval within which we are confident (with a certain percentage of confidence) the population parameter will fall

T	CI low	CI high
0.99	\$ 76,953	\$ 108,114

$$SE = \frac{s}{\sqrt{n}}$$

The **standard error** (often shortened to **SE**) is a number that tells us how much we expect a sample average to vary from the true average of the entire population.

The lower (or could be upper) bound of the mean

$$\bar{x} - \frac{s}{\sqrt{n}}$$

1. The General Word Formula

No matter which version you use, the basic idea is always:

$$\text{Confidence Interval} = \text{Point Estimate} \pm \text{Margin of Error}$$

- **Point Estimate:** This is the average you found from your sample (the group you actually measured).
- **Margin of Error:** This is the “plus or minus” amount that accounts for uncertainty.

2. The Math Formula (using the t-distribution)

$$CI = \bar{x} \pm \left(t^* \cdot \frac{s}{\sqrt{n}} \right)$$

- $\bar{x}$ (x-bar): The **mean** (average) of your sample.
- t^* : The **critical value**. This comes from a t-table based on how confident you want to be (like 95%) and your “degrees of freedom.”
- s : The **standard deviation** of your sample (how spread out the numbers are).
- n : The **sample size** (how many items or people you measured).
- $\frac{s}{\sqrt{n}}$: This whole part is called the **Standard Error**.